

Proof. Statement. Let $x \in [-1, \infty)$ and $n \in \mathbb{N}_0$. Then

$$(1+x)^n \geq 1+nx. \quad (*)$$

Equality holds precisely in one of the following situations

- $n = 0$ (any x);
- $n = 1$ (any x);
- $x = 0$ (any n).

1. Preliminaries. For a fixed integer $n \geq 0$ define

$$f_n(x) := (1+x)^n - 1 - nx.$$

Since f_n is a polynomial, all binomial coefficients $\binom{n}{k}$ are non-negative. We shall repeatedly use the elementary fact

$$x^2 \geq 0 \quad \text{for all } x \in \mathbb{R}. \quad (1)$$

2. Degenerate / edge cases. The inequality (??) is easy to verify in the following situations:

Case	^c Verification
$n = 0$	$(1+x)^0 = 1 = 1 + 0 \cdot x$
$n = 1$	$(1+x)^1 = 1 + x = 1 + 1 \cdot x$
$x = 0$	$(1+0)^n = 1 = 1 + n \cdot 0$
$x = -1, n \geq 1$	$(1-1)^n = 0 \leq 1 - n \leq 0$ (strict unless $n = 1$)

These cases will be excluded from the induction step; henceforth we assume $n \geq 2$ and $x \neq 0$.

3. Induction on n .

Base of induction. The inequality has already been proved for $n = 0, 1$ in the table above.

Induction hypothesis. Suppose that for a fixed $n \geq 2$

$$(1+x)^n \geq 1+nx \quad \text{for all } x \geq -1. \quad (\text{IH})$$

Inductive step. For $n+1$ we write

$$(1+x)^{n+1} = (1+x)^n(1+x) \stackrel{(\text{IH})}{\geq} (1+nx)(1+x). \quad (2)$$

Expanding the right-hand side gives

$$(1+nx)(1+x) = 1 + (n+1)x + nx^2. \quad (3)$$

Hence

$$(1+x)^{n+1} \geq 1 + (n+1)x + nx^2. \quad (4)$$

Since $n \geq 0$ and by (??) we have $nx^2 \geq 0$, (4) implies

$$(1+x)^{n+1} \geq 1 + (n+1)x. \quad (5)$$

Thus the statement holds for $n+1$. By the principle of mathematical induction, (??) is true for every integer $n \geq 0$ and all $x \geq -1$.

4. Equality cases. Equality in the final inequality (??) requires simultaneously

1. equality in the induction hypothesis, i.e. $(1+x)^n = 1 + nx$; and
2. equality in the non-negative term $nx^2 = 0$.

Condition2 forces either $n = 0$ or $x = 0$.

- If $n = 0$, we are in the base case where equality holds for every x .
- If $x = 0$, then $(1+0)^n = 1 = 1 + n \cdot 0$ for all n .

The remaining possibility is $n = 1$, which also yields equality for all x (see the edge-case table). No other choices satisfy both conditions.

Consequently, equality holds *iff* one of the three alternatives listed at the beginning of the proof occurs.

5. Conclusion. For every $x \geq -1$ and every integer $n \geq 0$,

$$\boxed{(1+x)^n \geq 1 + nx},$$

with equality exactly in the cases $n = 0$, $n = 1$, or $x = 0$. □